Entropy

The <u>entropy</u> of a discrete random variable x is a measure of the amount of information gained by the outcome of an experiment where we find the value of x.

Examples:

Let $x \in \{0,1\}$ represent a coin flip (heads=1, tails=0). To record the information about this coin flip we need 1-bit.

Now let $x \in \{0,1\}^n$ represent the results of *n* coin flips (with heads=1, tails=0 for each flip). To record the information about all *n* coin flips we need *n*-bits (i.e. a bit string of length *n*).

The <u>entropy</u> of x is a measure of the amount of information produced by flipping the coin. Or if the coin has not yet been flipped, it's a measure of the remaining uncertainty.

Definition:

Let x be a discrete random variable. The entropy of x is: $H(x) = -\sum P(x) \cdot \log_2\left[P(x)\right]$

Example1:

let $x \in \{00,01,10,11\}$ represent the results of *2* coin flips. Determine the entropy of x.

Each of these values has $P(x) = \frac{1}{4}$.

$$
\begin{aligned}
H(x) &= -\sum P(x) \cdot \log_2\left[P(x)\right] \\
&= -\frac{1}{4}\log_2\left[\frac{1}{4}\right] - \frac{1}{4}\log_2\left[\frac{1}{4}\right] - \frac{1}{4}\log_2\left[\frac{1}{4}\right] - \frac{1}{4}\log_2\left[\frac{1}{4}\right] \\
&= -\frac{1}{4}(-2) - \frac{1}{4}(-2) - \frac{1}{4}(-2) - \frac{1}{4}(-2) \\
&= \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} \\
&= 2
\end{aligned}
$$

Example2:

A cryptosystem consists of the following plaintext, ciphertext, and key sets:

$x \in M = \{a, b\}$, the set of all plaintext strings

$y \in C = \{1, 2, 3, 4\}$, the set of all ciphertext strings

$k \in K = \{k_1, k_2, k_3\}$, the set of all encryption keys

The probability distributions for *x* and *k* are: $P(a) = \frac{1}{4}, P(b) = \frac{3}{4}$ and $P(k_1) = \frac{1}{2}, P(k_2) = \frac{1}{4}, P(k_3) = \frac{1}{4}$

The encryption matrix is:

	a	*b*
k_1	1	2
k_2	2	3
k_3	3	4

Determine the amount of uncertainty in the key.

The random variable associated with the key is k so we find the entropy of k:

$$\begin{aligned}
H(k) &= -\sum P(k) \cdot \log_2\left[P(k)\right] \\
&= -P(k_1) \cdot \log_2\left[P(k_1)\right] - P(k_2) \cdot \log_2\left[P(k_2)\right] - P(k_3) \cdot \log_2\left[P(k_3)\right] \\
&= -\frac{1}{2}\log_2\left[\frac{1}{2}\right] \qquad -\frac{1}{4}\log_2\left[\frac{1}{4}\right] \qquad -\frac{1}{4}\log_2\left[\frac{1}{4}\right] \\
&= -\frac{1}{2}(-1) - \frac{1}{4}(-2) - \frac{1}{4}(-2) \\
&= \frac{1}{2} + \frac{1}{2} + \frac{1}{2} \\
&= 1.5
\end{aligned}$$

Exercises:

1. Let x represent the result of rolling a 6-sided dice. Determine the entropy of x.

2. Let x represent the result of rolling a "trick 6-sided dice" whose sides all show the same number: 3. Determine the entropy of x.

3. A pair of 4-sided dice is rolled. Let x represent if the roll is doubles (1 for doubles, 0 otherwise) and let y represent the sum on both dice.

a) Determine the entropy of x.

b) Determine the entropy of y.

4. A cryptosystem with plaintexts $x \in M = \{a, b\}$, ciphertexts $y \in C = \{1, 2, 3, 4\}$, and keys $k \in K = \{k_1, k_2, k_3\}$ has encryption matrix:

	a	*b*
k_1	1	2
k_2	2	3
k_3	3	4

And probability distributions: $P(a) = \frac{1}{4}, P(b) = \frac{3}{4}$ and $P(k_1) = \frac{1}{2}, P(k_2) = \frac{1}{4}, P(k_3) = \frac{1}{4}$

a) Determine the amount of uncertainty in the plaintext.

b) Determine the amount of uncertainty in the ciphertext.

Answers:

1. $H(x) \cong 2.5850$

2. $H(x) = 0$

3.

a) $H(x) \cong 0.8113$

b) $H(y) \cong 2.6556$

4.

a) $H(x) \cong 0.8113$

b) $H(y) \cong 1.8496$